Design Abstractions

17-423/723 Software System Design

Lecture 3
Jan 21, 2026

Learning Goals

- Describe the role of abstraction in communicating designs
- Describe common types of design abstractions
- Select and apply a suitable notation to model an aspect of a design
- Generate questions to explore further design decisions given a model

Logistics

- Homework 1 is out on Canvas; due next Wednesday, Jan 28
- Practice on building a context model (last lecture) & design models (today)
- Submission through Gradescope
- Reminder: Project team formation survey due the end of **<u>today</u>**
- **Recitation:** Using design models to understand a system

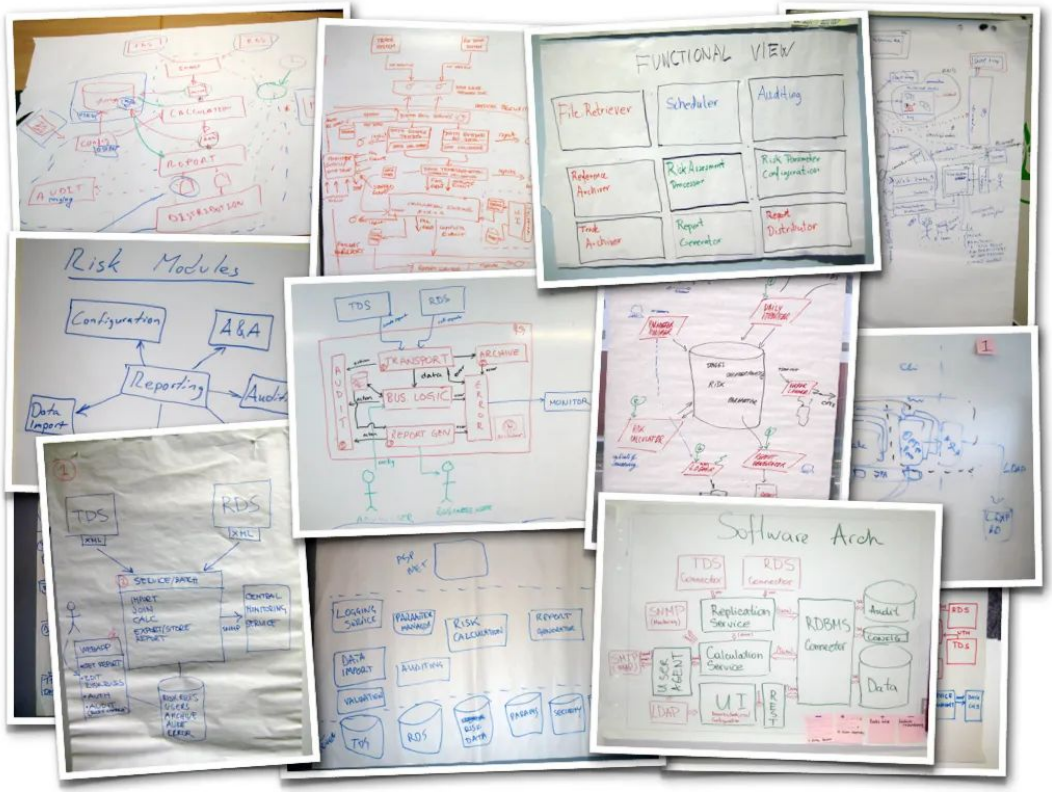
Documenting & Communicating Designs

Last class: Use of a context model to understand the **problem space** (domain entities & assumptions)

This class: Models for communicating design ideas in the **solution space**

How do software designers communicate ideas?

Designers use sketches

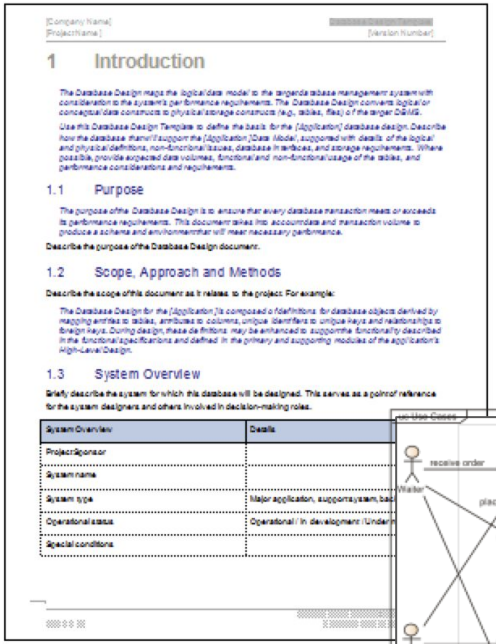


To brainstorm ideas

To explain how a design works

To ask questions about designs

Designers document their work


[Company Name]
[Project Name]
[Version Number]

1 Introduction

The Database Design maps the logical data model to the targeted database management system with consideration to the system's performance requirements. The Database Design converts logical or conceptual data constructs to physical storage constructs (e.g., tables, files) of the target DBMS.

Use this Database Design Template to define the basis for the [Application] database design. Describe how the database manor'll support the [Application] Data Model, supported with details of the logical and physical definitions, non-functional issues, database hardware, and storage requirements. Where possible, provide expected data volumes, functional and non-functional usage of the tables, and performance considerations and requirements.

1.1 Purpose

The purpose of the Database Design is to ensure that every database transaction meets or exceeds its performance requirements. This document takes into accounddata and transaction volume to produce a scheme and environmenthat will meet necessary performance.

Describe the purpose of the Database Design document.

1.2 Scope, Approach and Methods

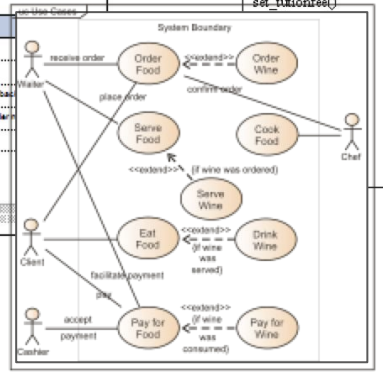
Describe the scope of this document as it relates to the project. For example:

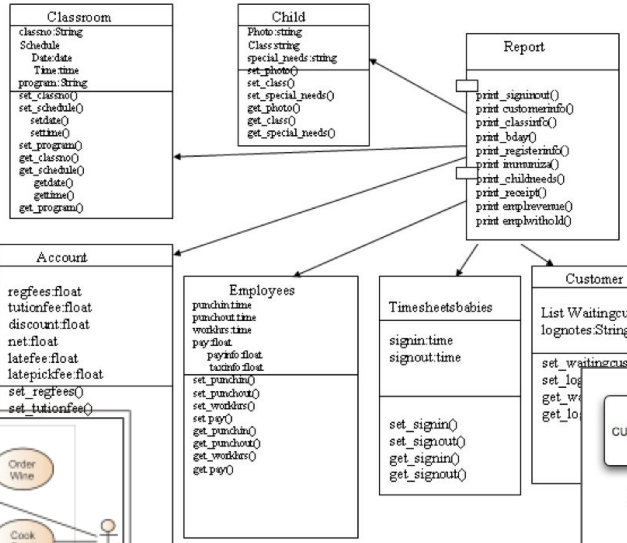
The Database Design for the [Application] is composed of definitions for database objects derived by mapping entities to tables, attributes to columns, unique identifiers to unique keys and relationships to foreign keys. During design, these definitions may be enhanced to support the functionality described in the functional specifications and defined in the primary and supporting modules of the application's High-Level Design.

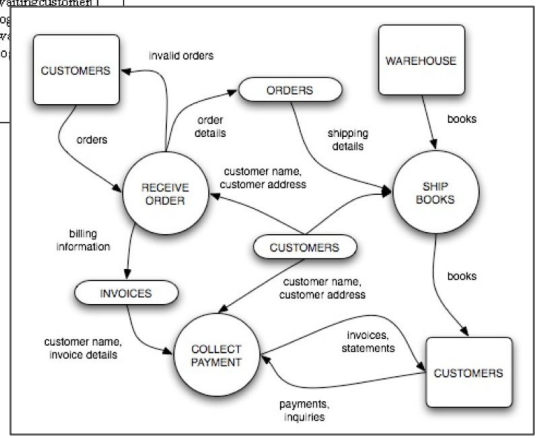
1.3 System Overview

Briefly describe the system for which this database will be designed. This serves as a point of reference for the system designers and others involved in decision-making roles.

System Overview	Details
Project Sponsor	
System name	
System type	Major application, supports system, bas
Operational status	Operational / In development / Under
Special conditions	







Viewpoint: Code as Design

Code is the final design and the only source of the truth

Source code listing is the only design documentation that we need

Q. What do you think? Arguments for & against?

Jack W. Reeves “Code as Design: Three Essays”
https://www.developerdotstar.com/mag/articles/reeves_design_main.html

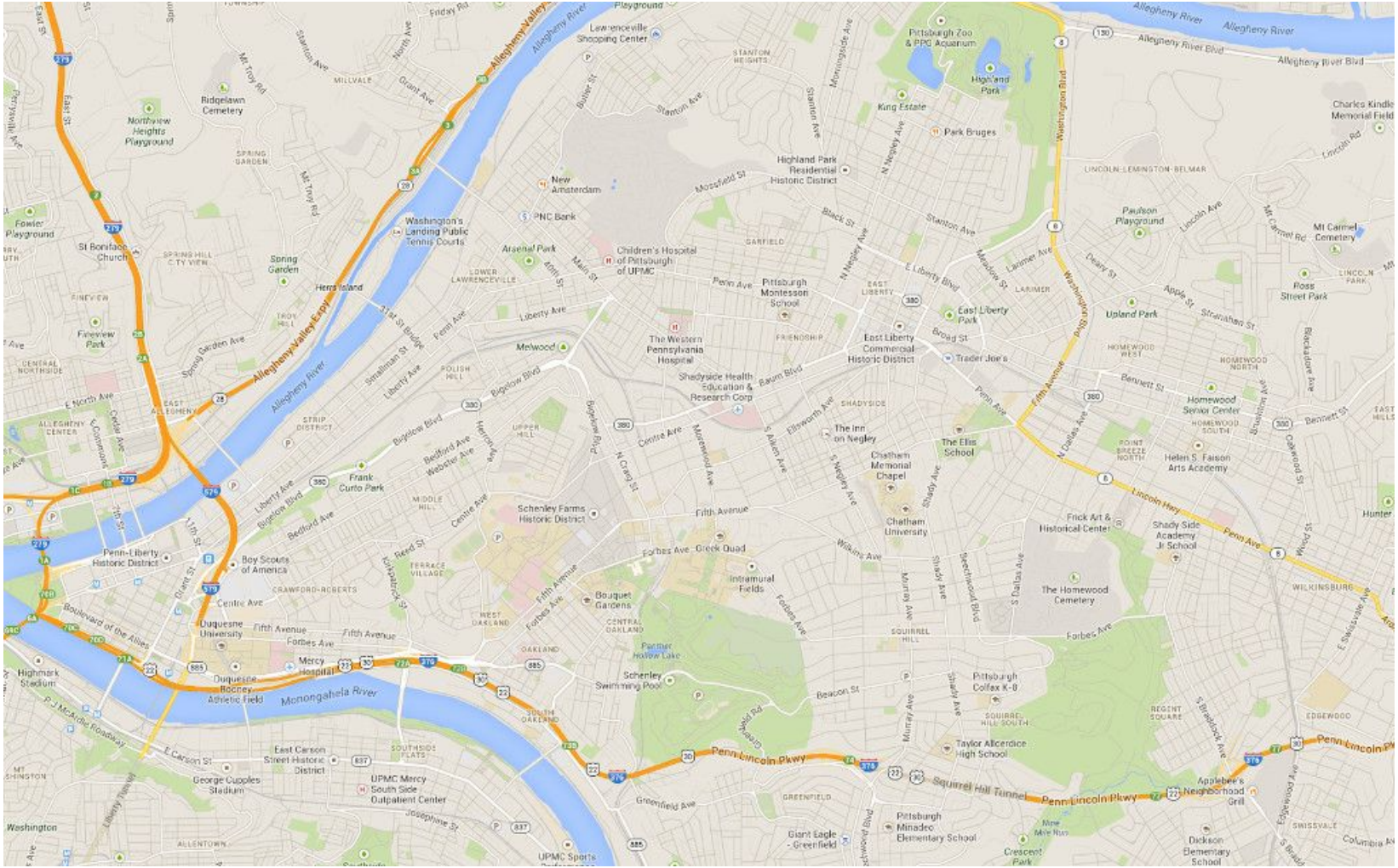
Designers use abstractions

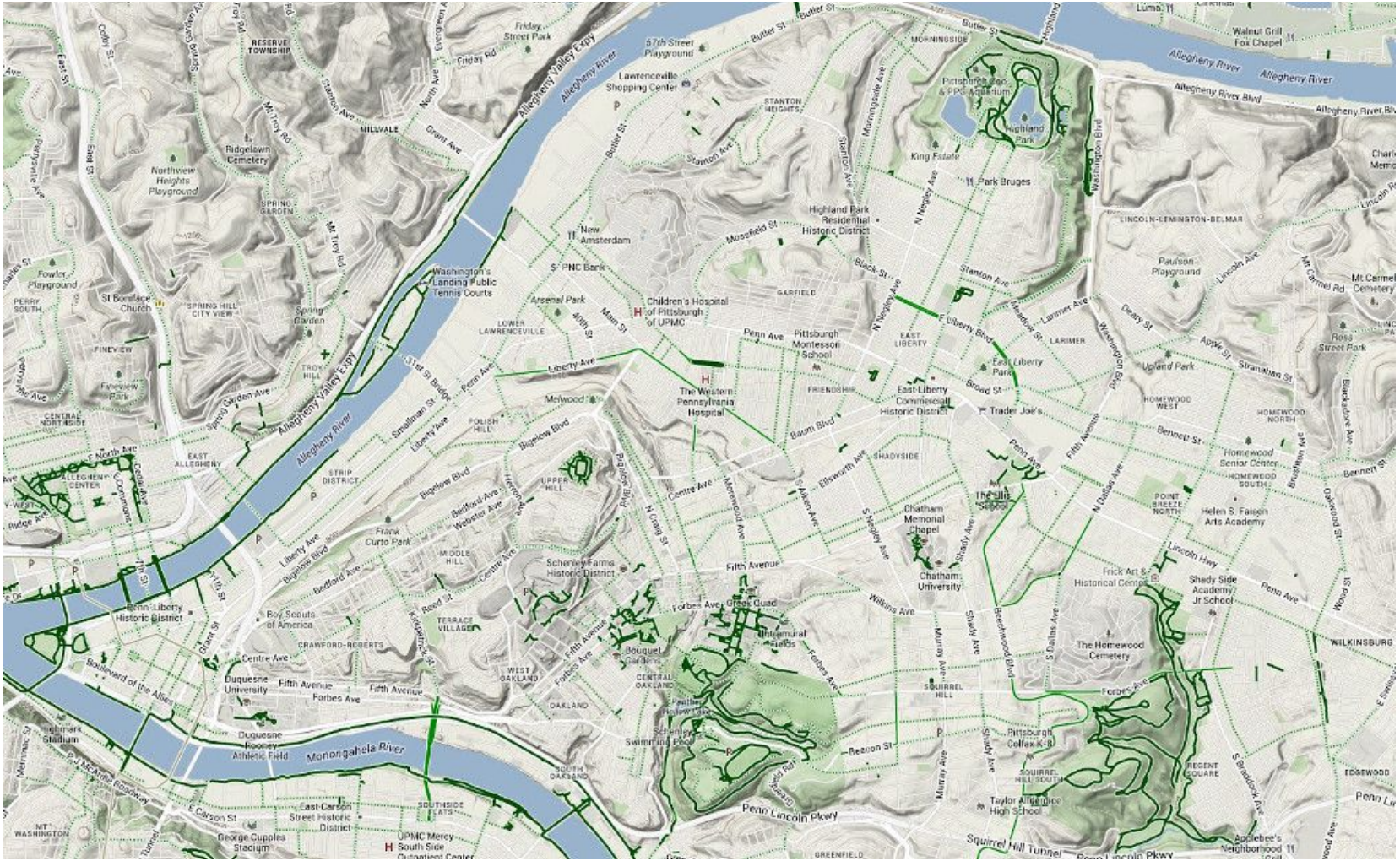
For most systems, code is too complex for a single person to understand

Code is not an ideal medium of communicating ideas

Code does not capture everything about design
- Q. What are some information that is not captured in the code?

Instead, designers communicate through **an abstraction of a system** – a description that focuses on a particular aspect & ignores other details


Allegheny River
Allegheny Valley Expy
Lawrenceville Shopping Center
Pittsburgh Zoo & PPG Aquarium
Highland Park
Allegheny River Blvd
Ridgelawn Cemetery
Northview Heights Playground
Mt Troy Rd
Stanton Ave
Grant Ave
Friday Rd
Butler St
Mossfield St
Highland Park Residential Historic District
King Estate
Park Bruges
Washington Blvd
Charles Kindle Memorial Field
Lincoln Rd
LINCOLN-LEMINGTON-BELMAR
Fowler Playground
St Boniface Church
Washington's Landing Public Tennis Courts
New Amsterdam
PNC Bank
Arsenal Park
Children's Hospital of Pittsburgh of UPMC
Black St
Paulson Playground
Mt Carmel Cemetery
Herrs Island
Spring Garden
Fineview Park
Main St
Penn Ave
Pittsburgh Montessori School
E Liberty Blvd
East Liberty Park
Upland Park
Ross Street Park
Liberty Ave
The Western Pennsylvania Hospital
East Liberty Commercial Historic District
Trader Joe's
Melwood
Shadyside Health Education & Research Corp
Baum Blvd
Bennett St
Homewood Senior Center
Bigelow Blvd
Centre Ave
The Inn on Negley
The Ellis School
Chatham Memorial Chapel
Chatham University
Frick Art & Historical Center
Shady Side Academy Jr School
Helen S. Faison Arts Academy
Lincoln Hwy
Penn Ave
Frank Curto Park
Bedford Ave
Schenley Farms Historic District
Fifth Avenue
Forbes Ave
Greek Quad
Intramural Fields
Penn-Liberty Historic District
Boy Scouts of America
Duquesne University
Mercy Hospital
Bouquet Gardens
Schenley Swimming Pool
The Homewood Cemetery
Pittsburgh Colfax K-8
Highmark Stadium
Duquesne Bocce Athletic Field
Monongahela River
P.J. McArdle Roadway
Boulevard of the Allies
Penn-Lincoln Pkwy
Squirrel Hill Tunnel
Taylor Allerdice High School
Applebee's Neighborhood Grill
East Carson Street Historic District
George Cupples Stadium
UPMC Mercy South Side Outpatient Center
Greenfield Ave
Giant Eagle - Greenfield
Pittsburgh Minadeo Elementary School
Dickson Elementary School
Crescent Park
Washington
UPMC Sports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Duquesne Rooney Athletic Field
Heinz Field
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Highland Park Residential Historic District
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Frick Art & Historical Center
Shady Side Academy Jr School
Pittsburgh Colfax K-8
Taylor Allderdice High School
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Centre Ave
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Wilkins Ave
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Baum Blvd
Ellsworth Ave
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Negley Ave
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Highland Park Bridge
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Squirrel Hill
Squirrel Hill South
Regent Square
Edgewood
Wilkinsburg
Oakland
Central Oakland
West Oakland
South Oakland
Greenfield
Southside Flats
Lower Lawrenceville
Garfield
East Liberty
Friendship
Larimer
Hillvale
Spring Garden
Spring Hill City View
Fineview
Troy Hill
Polish Hill
Strip District
Upper Hill
Middle Hill
Terrace Villages
Crawford-Roberts
Bloomfield
Morningside
Stanton Heights
Lincoln Park Zoo
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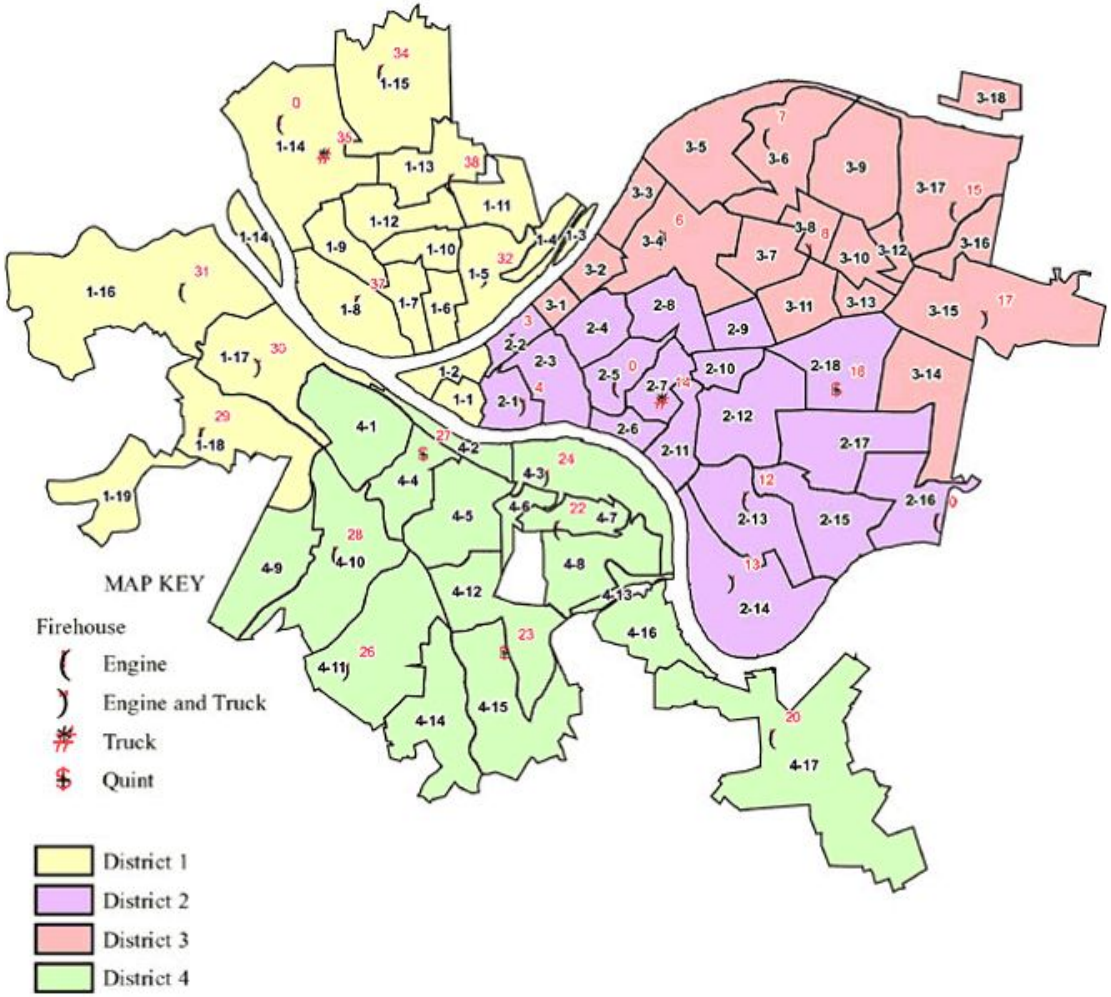

MAP KEY
Firehouse
Engine
Engine and Truck
Truck
Quint
District 1
District 2
District 3
District 4

Purpose of an Abstraction

Abstractions aid in understanding

Each abstraction highlights particular aspects of a system and deliberately hides other details

Abstractions facilitate reasoning

Each abstraction answers certain types of questions about the system

Abstractions are reusable

Each abstraction captures a commonality across multiple systems (within a single domain or sometimes across multiple domains)

What can we reason about?

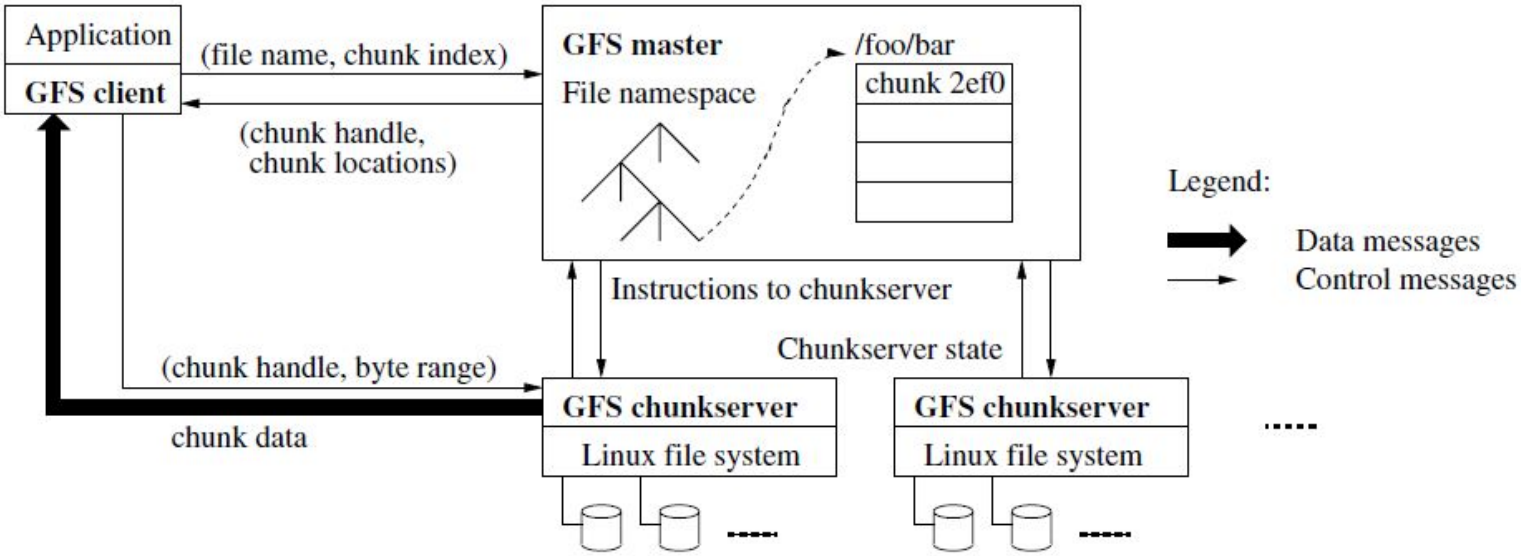
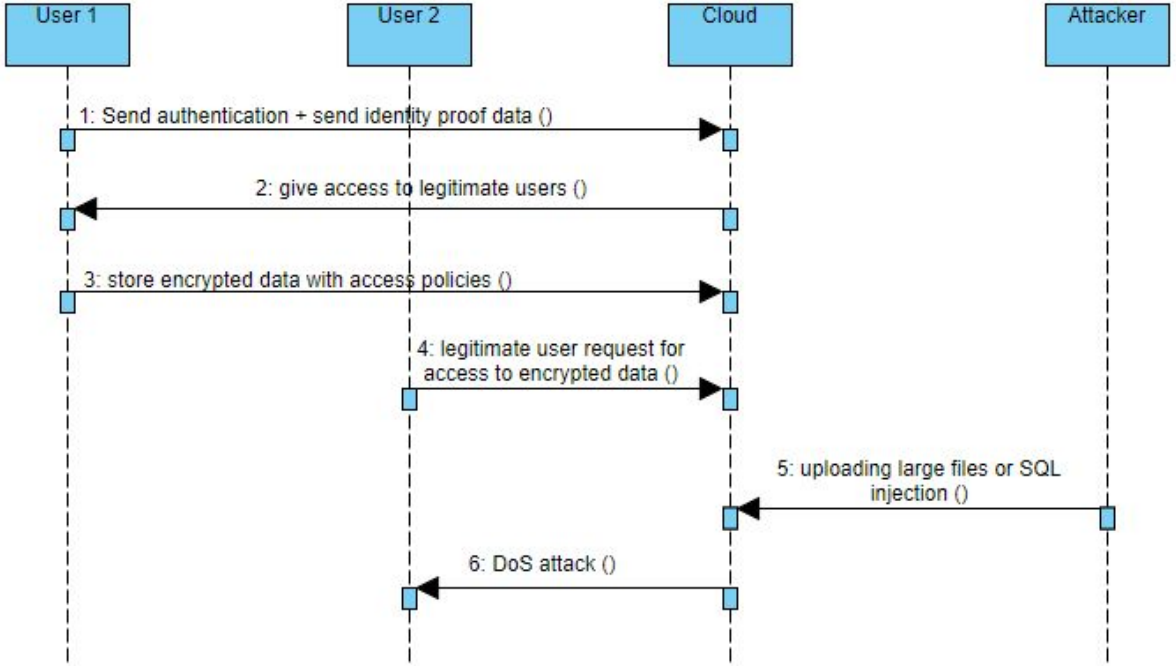


Figure 1: GFS Architecture

Ghemawat, Sanjay, Howard Gobioff, and Shun-Tak Leung. "The Google file system." ACM SIGOPS operating systems review. Vol. 37. No. 5. ACM, 2003.

What can we reason about?



Web of Things: Security Challenges and Mechanisms. Sardar & Anees. IEEE Access (2021).

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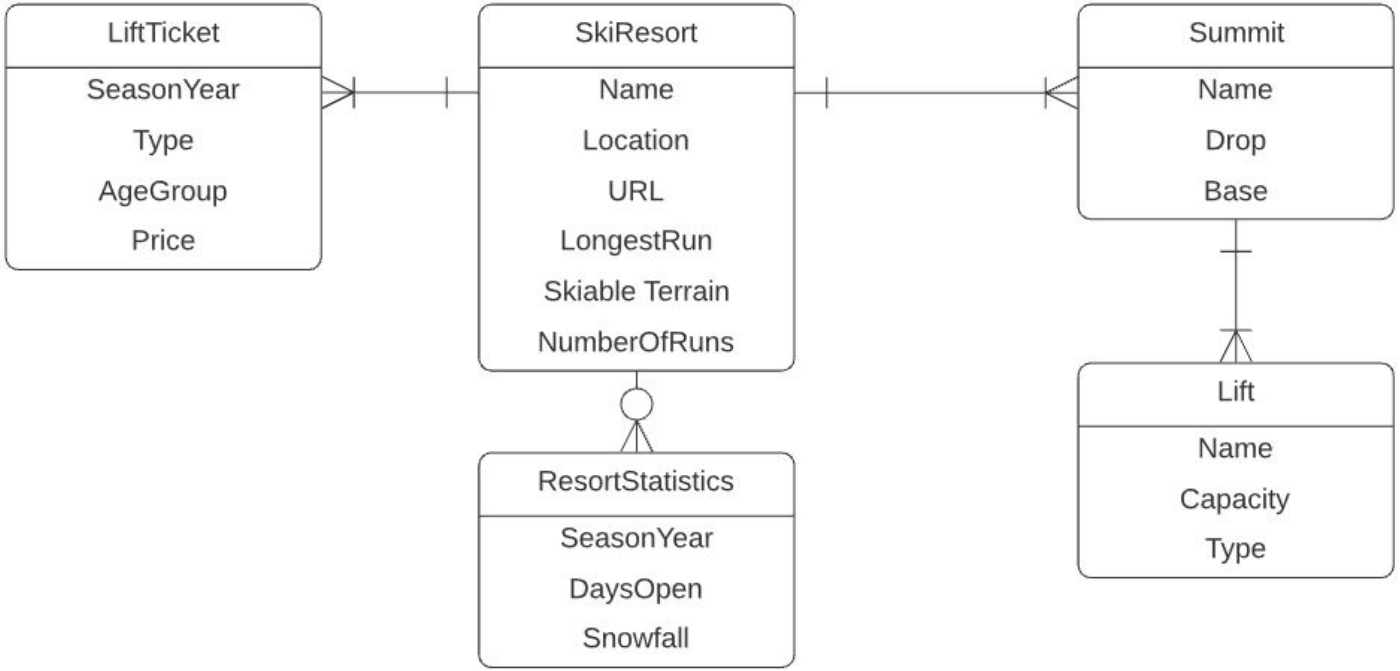


Notations for Common Design Abstractions

Component diagrams: What are major components in the system and how do they communicate with each other?

State machines: What states can the system be in, and what events cause it to change its state?

Data models: What types of data does the system store and what are their relationships?

Sequence diagrams: How different actors (domain entities & system components) collaborate to carry out some functionality?

Modeling in this class

Sometimes, models are themselves developed as a formal, first-class artifact

- To rigorously specify & verify properties about the system
- Completeness is an important property of the model in this case

In this class, the goal is mainly to **communicate**

- To explain a design to someone who will work on the system later (sometimes, yourself)
- To articulate an aspect of the system that is particularly complex
- To guide an AI coding agent towards a desirable implementation
- Clarity, rather than completeness, is more important

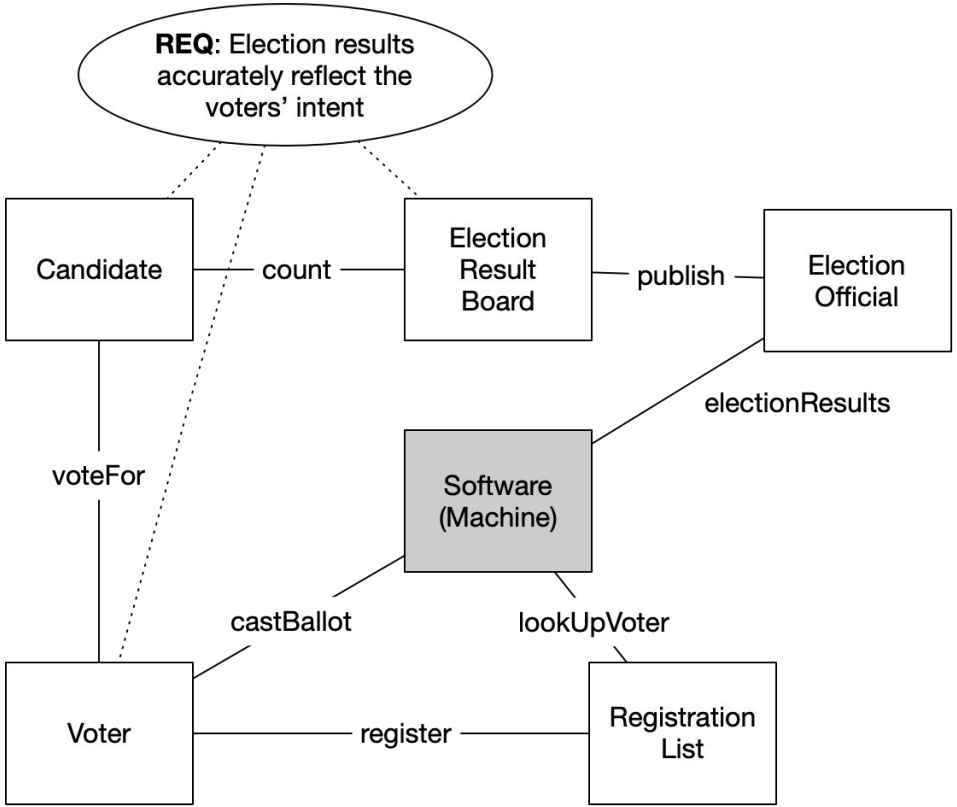
Example: Electronic Voting System

Voting System: Workflow

- A voter walks to an electronic voting machine and enters their voter ID information.
- The voting software checks whether the voter is registered to vote.
- The voter is presented with an electronic ballot with a list of candidates.
- The voter selects a candidate and casts their ballot by confirming their choice.
- The voting software collects all of the ballots and produces the final count for each candidate.
- The election officials retrieve and post the election results on a public board.

Q. What are some potential risks of failure with this system? What type of quality attributes should we design for?

Context Model for the Voting System



Component Diagram

Component Diagram

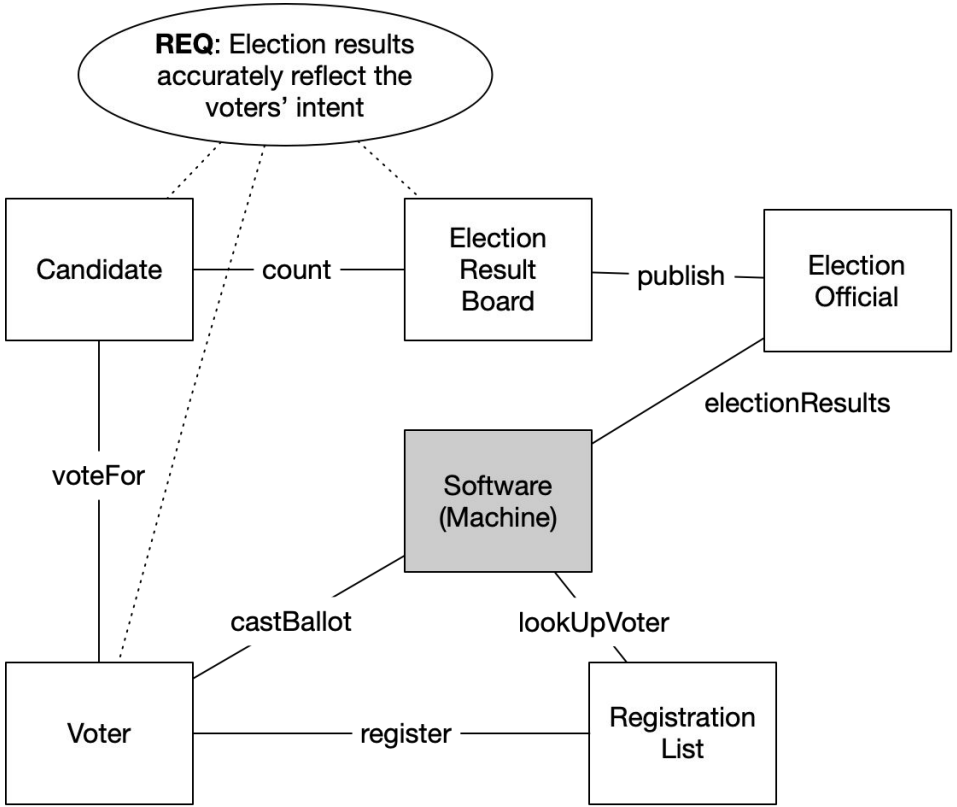
Purpose

Describe the major components in a software system, which components communicate with each other, and how they communicate.

Building blocks

- **Component**: A software component, responsible for carrying out a distinct unit of functionality
- **Connection**: A **directed** connection between a pair of components, labeled with an event (e.g., an API call) or data flow
- **Component responsibilities** (in text annotations): Describe the responsibilities of each component.

Designing the Voting Software: Decisions to Make



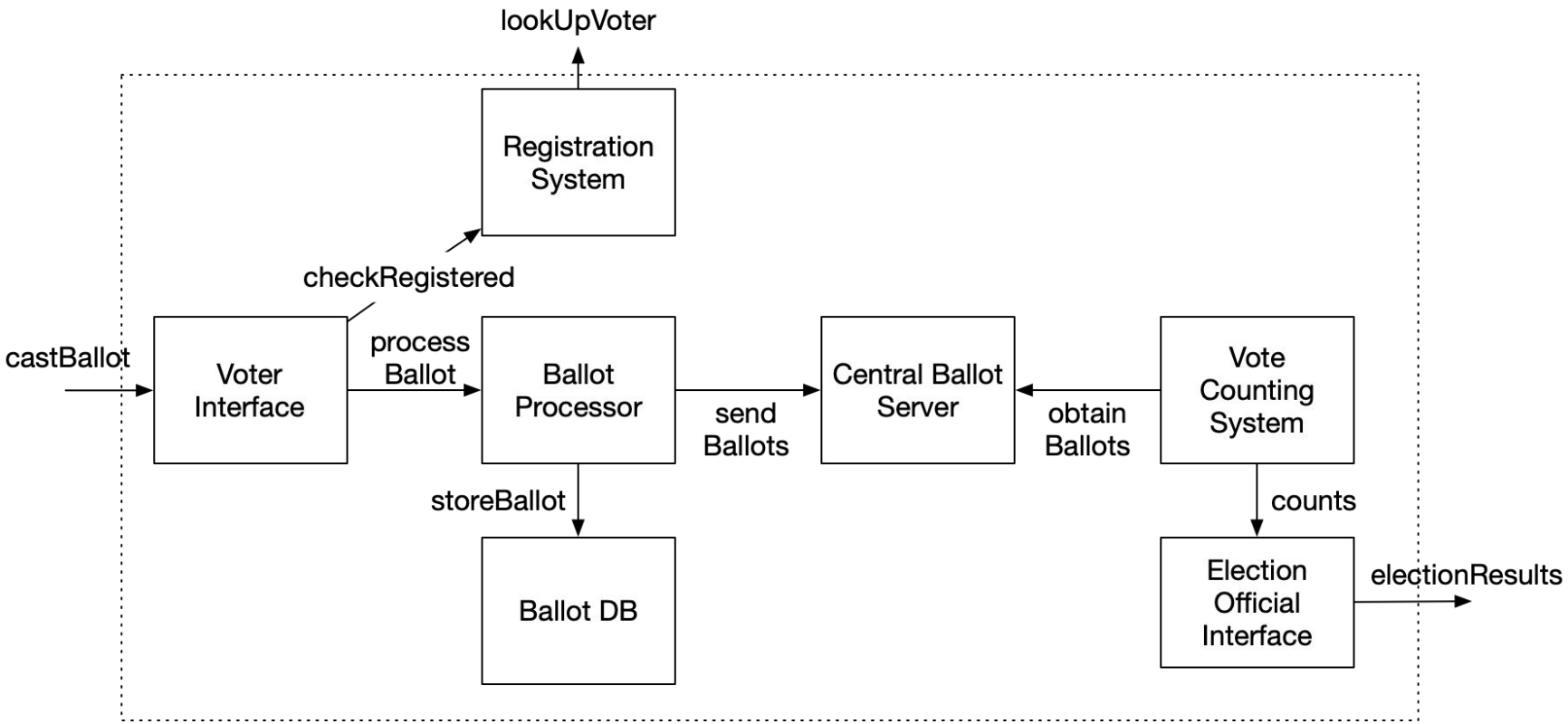
What is the specification of the voting software?

What components does the voting software consist of?

How do the components interact with the domain entities?

How do these components interact with each other?

Component Diagram: Example



Component Responsibilities

Voter Interface: Receives inputs from the voter & forwards ballot information to the Ballot Processor.

Registration System: Checks whether a voter is registered by looking up an external registration list.

Ballot Processor: Processes & stores each ballot on a local database. Uploads all of the ballots to the central server.

BallotDB: Stores the ballots cast so far for a local precinct.

Central Ballot Server: Stores all of the ballots across multiple precincts.

Vote Counting System: Compute the final tally of votes for each candidate.

Election Official Interface: Serves requests from an election official to retrieve election results.

Component Diagram: Questions to Ask

Each model is a starting point for asking & discussing further design questions

Different types of models encourage different types of questions

Component Diagram: Questions to Ask

What information is passed between one component to another? Is there a return value?

Which components interact with the entities in the problem space (e.g., users)?

Where is each component deployed? Which components are deployed on the same device?

What type of communication mechanism is used for each connection? (e.g., HTTP/S, RPC, Bluetooth) Is the communication synchronous or asynchronous?

What if a component changes or fails? What other components does it affect?

What if a connection fails? How does it affect the overall system?

State Machine

State Machine

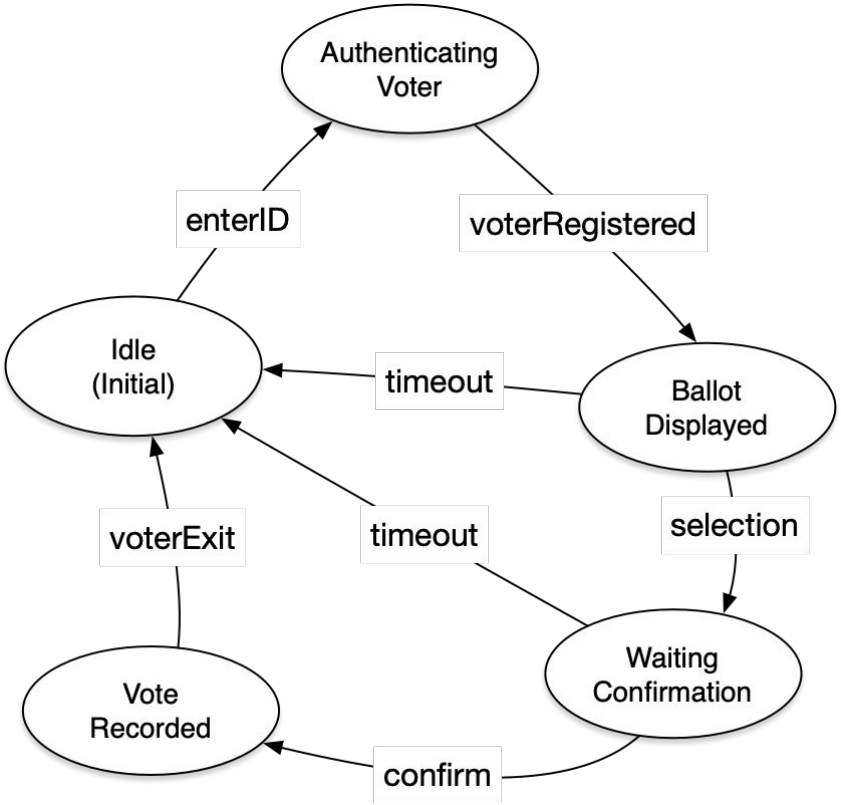
Purpose

- Describe possible states of a system (or its component) and which event triggers its state to change from one to another
- Often used to illustrate complex business logic or protocols

Building blocks

- **State:** A state of a system or a component
  - **Initial state:** A special state where the system/component begins its execution
- **Transition:** A change from one state to another based on an event

State Machine: Voter Interface



State Machine: Questions to Ask

What are possible error states? How can the machine reach those states?

Given a state, which events can occur? Does the state have a transition for each such event?

Can the machine get stuck in a state due to an event that never takes place?

Can it get into a cycle, and is that acceptable?

Is the machine deterministic; i.e., given a state and an event, is there at most one outgoing transition?

Data Model

Credit: Examples & figures by Daniel Jackson (MIT)

Data Model

Purpose

- Describe different types of data that the system needs to remember to fulfill its specification
- Serve as the conceptual schema for designing a database

Building blocks

- **Data type**: A collection of data elements or objects
- **Relation**: A **directed** relation between a pair of data types
- **Multiplicity constraints**: A constraint on a relation, specifying how many instances of the two data types can be related to each other

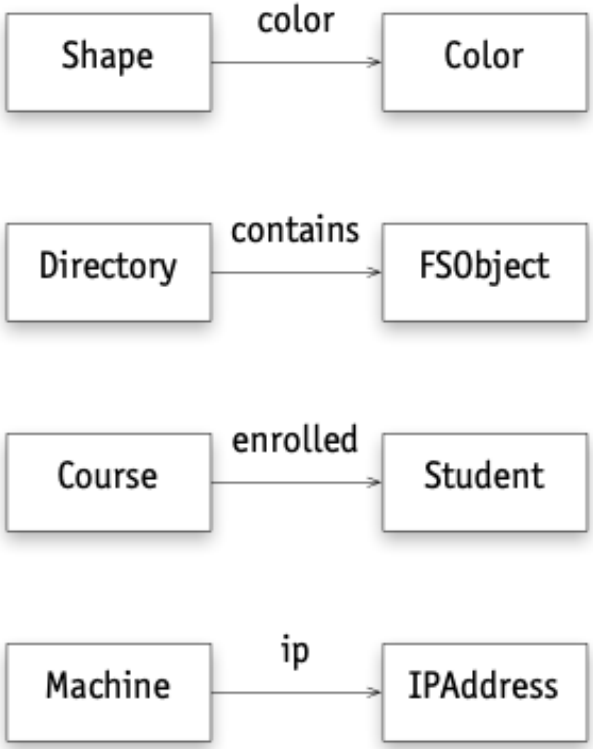
Data Model: Relations

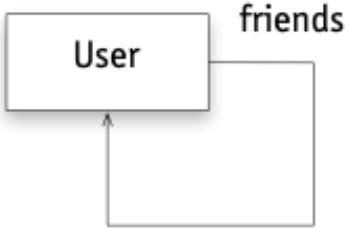
A relationship between different data types

Kinds of relations:

- Property
- Containment
- Association
- Naming

Q. What kinds of relations are these?

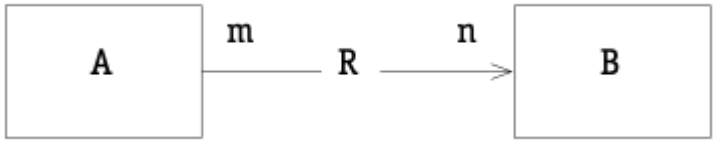




Data Model: Multiplicity Constraints

Constraints on relations

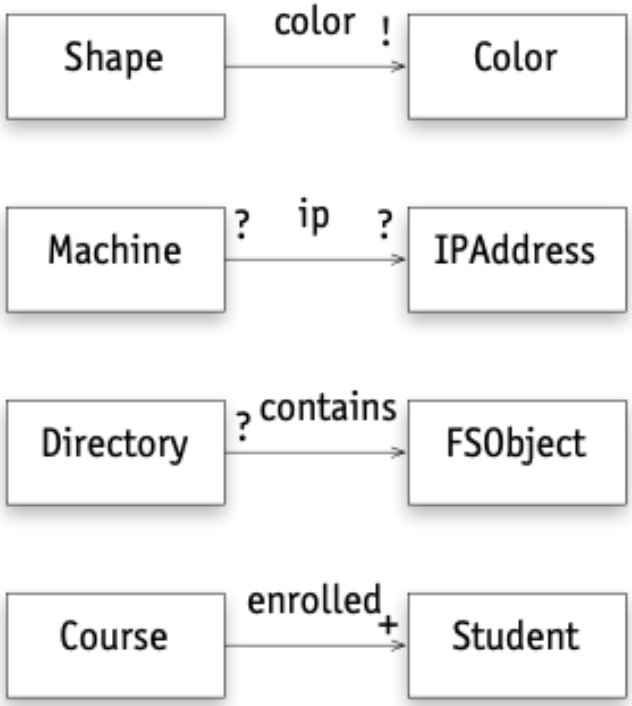
- R is a relation of type A to B
- R maps **m** A's to each B
- R maps each A to **n** B's

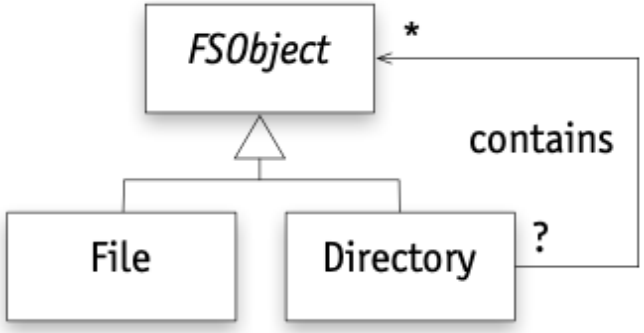


These constraints will eventually need to be enforced in the database design

+ one or more
* zero or more
! exactly one
? at most one
omitted = *

Data Model: Multiplicity Examples





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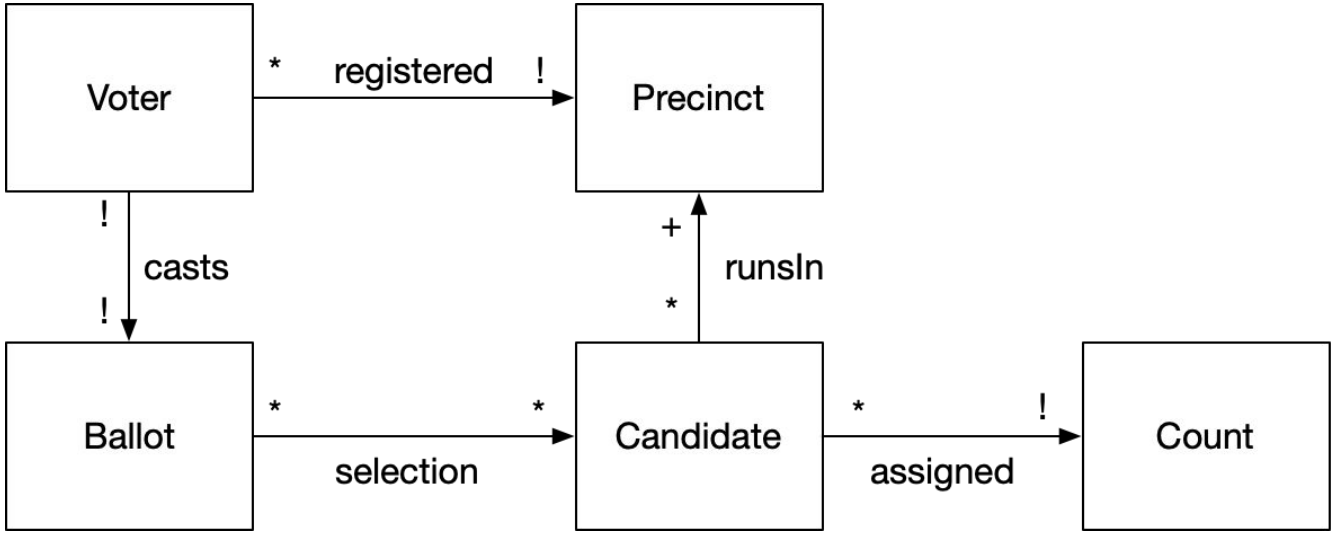
Data Model: Voting System



Data Model: Questions to Ask

Are we capturing all information that the system needs to function and achieve its quality attributes (e.g., security, availability)?

Do the multiplicity constraints reflect the real world scenarios? Are there any missing constraints? Are some constraints too strong?

Will all relations be stored on the same database, or be distributed across multiple databases? If distributed, do we need to consider consistency issues?

Is some of the data potentially sensitive? Do we need additional security or privacy mechanisms?

Sequence Diagram

Sequence Diagram

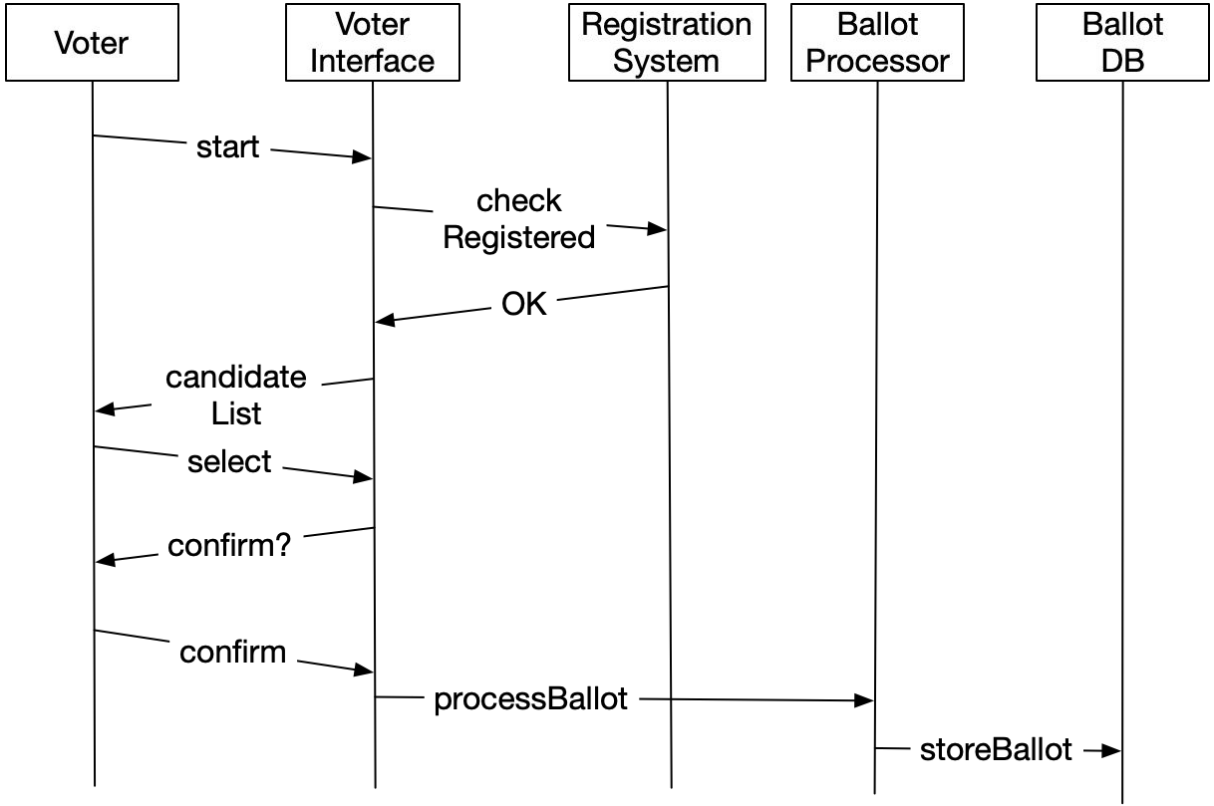
Purpose

Describe how a set of domain entities and system components collaborate in sequence to achieve a piece of functionality

Building blocks

- **Process**: A domain entity or system component
- **Message**: A message passed from one process to another

Sequence Diagram: Example



Sequence Diagram: Questions to Ask

What happens if a process terminates its activity early?

Is it possible for a message to be lost, and how does the system handle this? What if it arrives late?

What if a process receives two messages out of order? Does the order of execution matter for functionality or a quality attribute?

Are there any other processes that we are missing in this scenario?

Voter Fraud

FRANKFORT — A former Clay County precinct worker testified Friday that top election officers in the county taught her how to change people's choices on voting machines to steal votes in the May 2006 primary.

Voters walk away from the machine before pressing “confirm”
Election officials enter booth, press “back” & modify the vote

Tips for Building Diagrams

Aim for simplicity & clarity!

Keep diagrams to a reasonable size. If a diagram gets too big, break it into multiple ones.

Annotate a diagram with text to explain a concept (e.g., the responsibility of a component) if its meaning is not obvious from its label

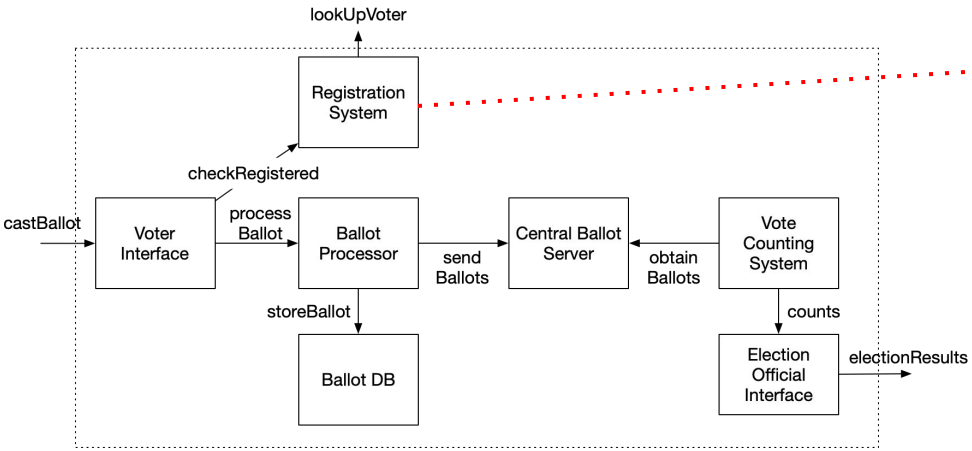
Use intuitive names! Avoid meaningless names (e.g., component named “Service”).

Consistency between Diagrams

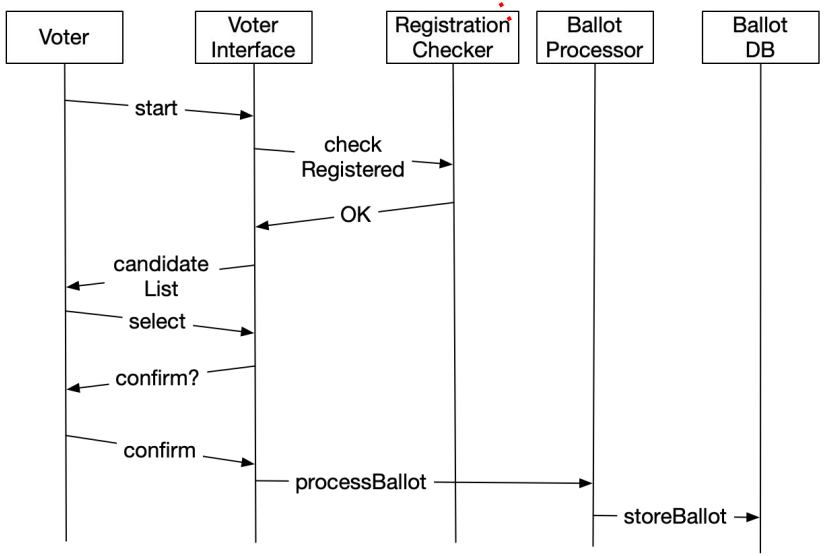
Try to use consistent names for the same concepts (components, events, etc.,) across multiple diagrams

A concept in one diagram may be represented by multiple concepts in another diagram; if so, explain how they are related

Consistency between Diagrams



Are these two the same?



Voter input "castBallot" is implemented by Voter Interface as a sequence of events "start", "select", and "confirm"

Tips for Building Diagrams

We will ask you describe your design using some of these models throughout project milestones

Use these models in your team discussions! We will ask for your reflections on how they helped (or not) with designing your system

Treat these models as a tool for brainstorming & communicating, not for documenting everything perfectly

- Focus on aspects that are most important for your system (recall “risk-driven” approach to design)
- Use models to ask further questions about the system!

Summary

Exit ticket!